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IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID: 25-26J-383

1. Topic (12 words max) -

Vocal Path -A Cloud-Enabled Intelligent System for Adaptive Speech Therapy in Stuttering

2. Research group the project belongs to –

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to –

Information Technology (IT)

4. If a continuation of a previous project: -

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Stuttering is a speech fluency disorder characterized by involuntary repetitions, prolongations, or blocks during speech, disrupting the normal rhythm and flow of verbal communication [1][2]. It commonly emerges between ages 5–10, a critical period for social and cognitive development, and can significantly affect children's confidence, classroom participation, and long-term psychosocial health [3][4]. Although stuttering does not reflect reduced intelligence, it often leads to anxiety, avoidance of communication, and social withdrawal [5][6]. Early diagnosis and timely intervention are therefore crucial to improving fluency and communicative outcomes.

Traditional diagnostic methods rely on subjective clinician assessments, which are time-intensive, inconsistent, and inaccessible in many regions due to limited speech-language pathologist (SLP) availability [2][4][7]. Automated speech analysis and machine learning (ML) models have shown great promise in identifying stuttering through acoustic and prosodic features such as Mel-Frequency Cepstral Coefficients (MFCCs), pitch variation, and pause duration [1][8][9][10]. However, most existing systems are developed for English-language datasets and lack adaptation to local linguistic contexts like Sinhala, reducing their generalizability. Additionally, current digital tools often provide static evaluations rather than dynamic, personalized feedback, limiting their use for continuous therapy and progress tracking [7][8][11].

Recent research emphasizes the integration of artificial intelligence (AI) and cloud computing in adaptive therapy environments [12][13][14]. Studies indicate that combining speech recognition with deep learning classifiers—such as CNN, RNN, and ConvLSTM—can improve detection accuracy and provide real-time stuttering assessment [8][15]. Yet, few systems incorporate culturally relevant, language-specific features or address the need for child-centered design in digital fluency therapy [16][17].

To address these limitations, the proposed study introduces **Vocal Path**, a cloud-enabled AI-driven system for adaptive speech therapy in stuttering. Vocal Path employs an audio-only ML classifier to detect disfluencies using acoustic and temporal speech patterns and delivers personalized pronunciation feedback with real-time adaptive exercises. The platform integrates parents, clinicians, and children in a shared digital ecosystem, allowing continuous progress tracking, data-driven feedback, and accessibility via mobile devices. Its emphasis on Sinhala-language support and AI-based fluency monitoring aims to revolutionize early diagnosis, scalable intervention, and long-term management of childhood stuttering in Sri Lanka.

References

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[5] Belal, M., Amer, A., & Abou-Elsaad, T. (2024). *Parent–Child Interaction Therapy for Children Who Stutter*. Egypt J. Otolaryngol., 40, 156.

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[7] Johnson, G. (2025). *Adapting the Lidcombe Program for 6–12-Year-Olds*. J. Fluency Disord.

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6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

Vocal Path is an AI-driven, cloud-based therapeutic platform designed to deliver adaptive, accessible, and personalized speech therapy for children who stutter. It addresses the limitations of conventional, clinician-dependent therapy by integrating automated detection, real-time feedback, and continuous progress tracking within a single, child-friendly mobile ecosystem. The system comprises four key components:

1. **ML-Based Stuttering Speech Classifier –**
   This module automatically identifies stuttering events such as sound repetitions, prolongations, and pauses by analyzing acoustic features (MFCCs, pitch, jitter). Machine learning models (e.g., CNN, Random Forest, SVM) are trained on labeled child-speech datasets and deployed through a cloud API (Firebase/Render) for real-time fluency assessment and continuous monitoring.
2. **Personalized Exercise Recommender –**
   Using hybrid rule-based and ML algorithms (e.g., KNN), this recommender adapts therapy exercises to the child's fluency progress and performance history. The system dynamically modifies session difficulty, mimicking clinician-guided personalization while supporting independent practice at home.
3. **Pronunciation Feedback System –**
   Powered by Automatic Speech Recognition (Google STT or Whisper), this module transcribes speech and evaluates it against target words using phoneme-level matching. It delivers immediate corrective feedback via an interactive mobile interface, promoting self-paced fluency improvement.
4. **Progress Tracking and Predictive Dashboard –**
   A cloud-based analytics module stores session data (scores, attempts, timestamps) in Firebase/MongoDB and visualizes fluency trends using Streamlit or Plotly Dash. Predictive models (e.g., Prophet) forecast progress and identify improvement plateaus, providing data-driven insights for parents and therapists.

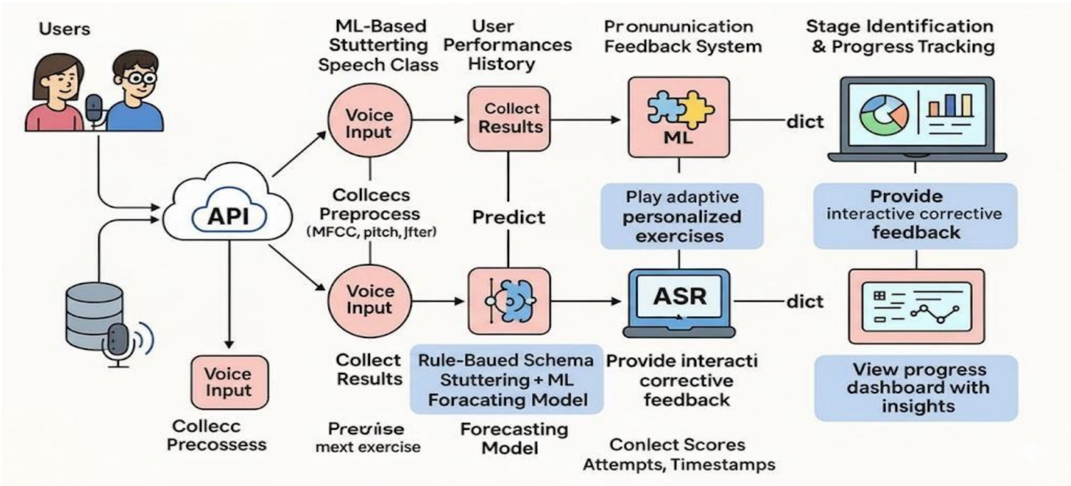

Figure 1 conceptual diagram

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

1. Domain Expertise (Clinical/SLP Perspective)
This project requires specialized expertise in **speech-language pathology**, particularly in the diagnosis, evaluation, and treatment of **stuttering and other fluency disorders**. Certified **Speech-Language Pathologists (SLPs)** play a central role in defining stuttering markers such as repetitions, prolongations, and blocks, labeling datasets (stuttered vs. fluent speech), and validating the system's detection algorithms and therapy modules. Their clinical insights ensure that the exercises, assessment protocols, and feedback mechanisms align with **evidence-based stuttering intervention methods**, including behavioral fluency shaping and real-time feedback therapy.

Personnel Involved: Certified SLPs, Clinical Linguists, Child Speech Therapists, and Neurodevelopmental Specialists.

2. Technical Knowledge (AI, HCI, Software)
The **Vocal Path** platform integrates knowledge from **machine learning, digital signal processing, natural language processing (NLP), and human-computer interaction (HCI)**. ML engineers design and train classifiers using acoustic features such as **MFCCs, jitter, pitch, and pause duration** to distinguish fluent from disfluent speech. ASR developers implement **speech-to-text transcription** modules (e.g., Google STT or Whisper) for pronunciation feedback. Cloud developers handle system deployment via **Firebase, Vercel, and Render**, ensuring scalability and security. HCI and UX specialists design a **child-friendly, gamified interface** that supports motivation and accessibility for children and parents.

Personnel Involved: ML Engineers, ASR Developers, Software Engineers, UX/HCI Designers, and Cloud DevOps Experts.

3. Data Requirements (Speech and User Metrics)
The project requires a curated dataset of **child speech samples**—both stuttered and fluent—collected under SLP supervision. Data may come from open-access fluency corpora or ethically approved recordings from local clinics. Longitudinal user data (e.g., fluency scores, session timestamps, performance metrics) will support progress tracking and predictive modeling. All data handling adheres to **strict privacy regulations** such as **GDPR and HIPAA**, ensuring confidentiality and ethical compliance.

Personnel Involved: Data Scientists, Speech Data Curators, Clinical Data Managers, and Ethics/Compliance Officers.

8. Objectives and Novelty

Main Objective

The primary objective of this research is to develop **VocalPath**, a cloud-enabled AI-driven platform that provides personalized and adaptive speech-fluency therapy for children who stutter. The system uses machine learning, speech recognition, and real-time feedback to detect disfluencies, recommend targeted exercises, and track long-term progress. Designed for both independent home use and remote clinician supervision, it bridges the gap between clinic-based stuttering therapy and accessible at-home intervention.

Member Name with Registration No	Sub Objective	Tasks	Novelty
K.P.S Chithakshana IT22565044	ML-Based Stuttering Speech Classifier	**T1.1**: Collect and preprocess audio datasets (stuttering & typical speech). **T1.2**: Extract acoustic features (MFCCs, pitch, jitter, energy). **T1.3**: Train multiple ML models (SVM, RF, CNN) to classify disfluencies. **T1.4**: Optimize model using cross-validation. **T1.5**: Deploy the model as a Cloud/REST API. **T1.6**: Validate classifier accuracy using SLT-labeled samples.	Unlike systems relying on transcription or visual cues, this classifier uses **audio-only acoustic cues** to detect stuttering in real time. A **cloud-deployed API** enables scalable remote screening.

S.M.C.N Serasinghe IT22590862	Personalized Exercise Recommender	**T2.1**: Define rule-based schemas for exercise difficulty. **T2.2**: Build a performance database (scores, attempts, timestamps). **T2.3**: Train ML models (KNN/Decision Tree) to learn user patterns. **T2.4**: Combine rule-based + ML logic for adaptive therapy. **T2.5**: Build UI for selecting, rating & progressing through exercises. **T2.6**: Evaluate recommender accuracy through expert review.	Standard therapy apps are static. This system uniquely combines **rule-based scaffolding** with **adaptive ML** to offer **real-time, user-specific therapy paths** that evolve with performance.
R.G.S.N.C.Rathninda IT22563064	Feedback System for Pronunciation	**T3.1**: Integrate ASR (Google/Whisper) into frontend. **T3.2**: Implement string/phoneme-level scoring to compare produced vs target words. **T3.3**: Build a React frontend with recording + visual feedback. **T3.4**: Optimize latency and scoring feedback. **T3.5**: Conduct user tests to refine usability and accuracy.	Unlike Sinhala-based apps with only translation features, this module gives **live corrective feedback** (phoneme-level) tailored for children—filling a major clinical gap in digital therapy.
K.A.I.Fernando IT22273758	Identifying the stage of Stuttering+ Progress Tracking with Prediction Dashboard	**T4.1**: Build severity-grading model (mild/moderate/severe). **T4.2**: Design database structure for logs, scores, timestamps. **T4.3**: Implement dashboard using Streamlit/Plotly/Dash.	A **standardized framework** for Sinhala stuttering severity prediction. Uses **time-series forecasting**—something missing in current

		T4.4: Train time-series model (Prophet) for improvement forecasting. **T4.5**: Validate forecasting accuracy. **T4.6**: Generate exportable clinical reports. **T4.7**: Ensure data security and compliance.	tools—to predict future fluency improvement and assist clinicians with long-term planning.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
K.P.S Chithakshana IT22565044	**1. ML-Based Stuttering Speech Classifier** This component aligns with **computational speech pathology**, **machine learning**, and **signal processing**, where classification of disordered speech is a recognized area. MFCCs, pitch, jitter and energy extraction follow standard practices in speech analysis. Training SVM, RF and CNN models on clinical datasets complies with best practices in **pattern recognition** for speech disorders. Deployment as a REST API supports modern, **real-time diagnostic systems**.
S.M.C.N Serasinghe IT22590862	**2. Personalized Exercise Recommender** This component merges **clinical rehabilitation protocols** with **adaptive ML**. Rule-based scaffolding reflects speech therapy principles, while ML personalization (KNN/R-Tree) supports intelligent tutoring system design. It aligns with specialization areas such as **health informatics**, **AI-based personalization**, and **digital therapy systems**.
R.G.S.N.C.Rathninda IT22563064	**3. Feedback System for Pronunciation** This component applies knowledge in **ASR**, **phoneme-level analysis**, and **HCI**. Real-time transcription-based feedback aligns with **computer-assisted pronunciation training (CAPT)**. The interface follows **HCI accessibility** principles for therapeutic environments.

K.A.I.Fernando IT22273758	**4. Identifying the stage of Stuttering and Progress Tracking with Prediction Dashboard** This component aligns with **data analytics**, **forecasting models**, **visualization**, and **rehabilitation informatics**. Uses acoustic-based severity classification and **Prophet time-series forecasting** to model improvement trends. Visualization dashboards adhere to **health data interpretation** standards.

10 Supervisor details -

	Title	First Name	Last Name	Signature
Supervisor	Senior professor	Samantha	Thelijjagoda	
Co-Supervisor	Senior Lecturer	Junius	Anjana	
External Supervisor	Professor	Jithangi	Wanigasinghe	
Summary of external supervisor's (if any) experience and expertise				

<u>This part is to be filled by the Topic Screening Staff members.</u>

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation
and Recommendation for the Research topic:

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	**Signature**

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.